

ECON 21030: Honors Econometrics

Midterm 1

Spring Quarter 2023

1: [30 points] True, false, or it depends? If true, prove it or give an appropriate reference (briefly). If false, explain why or provide a counterexample. If it depends, explain why. No credit will be given without some justification, and good justification will be awarded full credit.

1. [5 points] If $Cov(X, Y) = Cov(Y, Z) = 0$, then $Cov(X, Z) = 0$.

This is false. Suppose it were true for X, Y, Z . Let $\tilde{X} = X + W$ and $\tilde{Z} = Z + W$ with $Cov(W, Y) = 0$ and $Var(W) > 0$. Then clearly, $Cov(\tilde{X}, Y) = Cov(Y, \tilde{Z}) = 0$, but $Cov(\tilde{X}, \tilde{Z}) = Var(W)$, so we have constructed a simple counter example.

2. [5 points] The law of iterated expectations implies that $E[Y] = E[E[Y|X]] = E[E[Y|\log(X)]]$.

True, so long as the conditional expectation of Y given X and Y given $Z = \log(X)$ are well defined. LIE simply holds regardless of what we condition on in the inner expectation.

3. [5 points] For an iid sample (X_i, Y_i) of size N where $(X_i, Y_i) \sim N(\mu, \Sigma)$, $Var(\bar{X}_N - \bar{Y}_N) = \frac{1}{N}(\sigma_X^2 + \sigma_Y^2)$, where $\bar{X}_N$ is the sample mean of X_i and σ_X^2 is the variance of X_i (and analogously for Y_i).

True if $Cov(X_i, Y_i) = 0$, otherwise we are missing a term since:

$$\begin{aligned} Var(\bar{X} - \bar{Y}) &= Var\left(\frac{1}{N} \sum_{i=1}^N X_i - Y_i\right) \\ &= \frac{1}{N} Var(X_i - Y_i) = \frac{1}{N} (\sigma_X^2 + \sigma_Y^2 - 2\rho_{XY}\sigma_X\sigma_Y) \end{aligned}$$

4. [5 points] A level- α t -test of H_0 will reject with probability exactly equal to α when H_0 is true.

It depends. This is true if the test-static has a finite-sample t distribution, as would be the case for hypotheses tests about the sample mean for iid observations from a normal distribution. But in general, the finite sample distribution need not be t even though we are constructing a t -test, and hence the level- α test may have a size different than α .

5. [5 points] An unbiased estimator of a given parameter is always the best choice.

It depends. We showed in Homework 1, for example, that for a set of iid observations from a uniform distribution over $[0, \theta]$ the sample max is biased for θ , but making it unbiased by adjusting by $\frac{n+1}{n}$ increases the sampling variance. Whether something is “best” depends on

how you benchmark performance and weigh bias versus variance. Minimizing MSE is one way to make that trade off.

6. [5 points] For a size N iid sample from a distribution with mean μ and variance $\sigma^2 < \infty$, then $E[\frac{1}{N} \sum_{i=1}^N X_i^2] = \mu^2$.

False, since this expectation evaluates to $E[X_i^2] = \sigma^2 + \mu^2$.

- 2: [25 points] Consider an iid sample $\{X_i\}_{i=1}^N$ from a uniform distribution over $[0, 1]$.

1. [5 points] Find a constant a such that $\bar{X}_N^2 = \frac{1}{N} \sum_{i=1}^N X_i^2 \xrightarrow{P} a$ and show that convergence in probability holds.

In this case, we know that: $E[X_i^2] = \int_0^1 x^2 dx = \frac{1}{3}$. So by WWL $\bar{X}_N^2$ will converge in probability whenever X_i^2 has finite variance, which is true because $Var(X_i^2) = \int_0^1 x^4 dx - \frac{1}{9} = \frac{1}{5} - \frac{1}{9}$.

2. [5 points] Find the limiting distribution of $\sqrt{N}(\bar{X}_N^2 - a)$.

The central limit theorem will apply here, so we know that $\sqrt{N}(\bar{X}_N^2 - a) \xrightarrow{D} N(0, Var(X_i^2)) = N(0, 4/45)$.

3. [5 points] Find the probability limit of $\log(\bar{X}_N^2)$.

Since we are dealing with a uniform distribution and $\bar{X}_N^2 > 0$ almost surely, continuous mapping theorem tells us that $\log(\bar{X}_N^2) \xrightarrow{P} \log(a) = -\log(3)$.

4. [5 points] Find the limiting distribution of $\sqrt{N}(\log(\bar{X}_N^2) - \log(a))$.

We can use the delta method here to show that $\sqrt{N}(\log(\bar{X}_N^2) - \log(a)) \xrightarrow{D} N(0, 9 \cdot 4/45) = N(0, 4/5)$.

5. [5 points] What is the length of a 95% confidence interval based on this approximation as a function of N ?

Using this approximation, the sampling distribution is normal with variance $4/(5N)$. To construct a 95% confidence interval using a z-score, we would simply use $2 \cdot 1.96$ times the standard error, which is $2/\sqrt{5N}$.

- 3: [25 points] Consider a model where individuals are endowed with productivity P_i . They can choose to acquire schooling, which costs C_i . Their wages are given by $Y_i = P_i + \beta S_i$, where S_i is a binary indicator for whether they acquired schooling.

1. [5 points] Assume individuals acquire schooling if the benefits in terms of wages outweigh the costs. Write down S_i as a function of C_i and P_i .

$$S_i = 1\{P_i + \beta - P_i - C_i > 0\} = 1\{\beta - C_i > 0\}$$

2. [5 points] Write down an expression for average wages conditional on $S_i = 1$ and $S_i = 0$ in terms of conditional moments of P_i .

$E[Y_i|S_i = 1] = \beta + E[P_i|S_i = 1]$ and $E[Y_i|S_i = 0] = E[P_i|S_i = 0]$. Since $S_i = 1\{\beta - C_i > 0\}$, these can also be expressed as $\beta + E[P_i|\beta > C_i]$ and $E[P_i|\beta \leq C_i]$.

3. [5 points] Under what conditions will the difference in conditional means identify β ?

One sufficient condition is that P_i is independent of C_i , so that e.g., $E[P_i|1\{\beta > C_i\}] = E[P_i]$.

4. [10 points] Suppose that P_i and C_i are jointly normally distributed with marginal means μ_P and μ_C , correlation ρ and marginal variances of 1. Derive an expression for the mean of Y_i given $S_i = 1$ and $S_i = 0$ in terms of the marginal means of P_i and C_i , their correlation, and properties of the normal. How does the difference between β and the difference in conditional means depend on ρ ? (hint: you may need the fact that for a standard normal pdf, $-\phi'(x) = x\phi(x)$)

From the properties of normals, we know that $E[P_i|C_i = c] = \mu_P + \rho(c - \mu_C)$. Thus:

$$\begin{aligned} E[P_i|\beta > C_i] &= E[E[P_i|\beta > C_i, C_i]|\beta > C_i] \\ &= E[\mu_P + \rho(C_i - \mu_C)|\beta > C_i] \\ &= \mu_P + \rho(E[C_i|\beta > C_i] - \mu_C) \end{aligned}$$

At this point we need to evaluate $E[C_i|\beta > C_i] = \int_{-\infty}^{\beta} c\phi(c - \mu_C)dc / \Phi(\beta - \mu_C)$. You can either apply the inverse Mills ratio formula directly, or using the hint, note that $\int_{-\infty}^{\beta} c\phi(c - \mu_C)dc = \int_{-\infty}^{\beta} -\phi'(c - \mu_C)dc = -\phi(\beta - \mu_C) + K$. The constant term K must equal μ_C , so that $E[C_i|\infty > C_i] = \mu_C$. So the final result is that:

$$E[Y_i|S_i = 1] = \beta + \mu_P + \rho \left(\frac{-\phi(\beta - \mu_C)}{\Phi(\beta - \mu_C)} \right)$$

Very similar logic will show you that:

$$E[Y_i|S_i = 0] = \mu_P + \rho \left(\frac{\phi(\beta - \mu_C)}{1 - \Phi(\beta - \mu_C)} \right)$$

If ρ is positive, you can see that the difference will be smaller than β , and the opposite if ρ is negative.

4. [40 points] Suppose you are interested in the following population regression:

$$Y_i = \alpha + \beta_1 X_i + \beta_2 W_i + \epsilon_i$$

However, you only observe the joint distribution of $(Y_i, X_i, \tilde{W}_i)$, where $\tilde{W}_i = W_i + u_i$, $u_i \sim N(0, \sigma_u^2)$ is an error-ridden version of W_i , with u_i independent of all other random variables. Hence you can only compute this alternative regression:

$$Y_i = \tilde{\alpha} + \tilde{\beta}_1 X_i + \tilde{\beta}_2 \tilde{W}_i + \tilde{\epsilon}_i$$

Your ultimate goal is to learn β_1 , which is core to your model. β_2 is not directly of interest.

1. [5 points] Derive an expression that relates β_1 to $\tilde{\beta}_1$ when $Cov(X_i, W_i) = 0$.

When $Cov(X_i, W_i) = 0$, β_1 is also the coefficient on X_i in the population regression of Y_i on X_i alone. As is $\tilde{\beta}_1$, since $Cov(X_i, \tilde{W}_i) = Cov(X_i, W_i + u_i) = Cov(X_i, W_i) = 0$. Hence $\beta_1 = \tilde{\beta}_1$.

2. [5 points] Derive an expression that relates β_2 to $\tilde{\beta}_2$ when $Cov(X_i, W_i) = 0$. What is the role of σ_u^2 ?

When $Cov(X_i, W_i) = 0$, β_2 is also the coefficient on W_i in the population regression of Y_i on W_i alone, and $\tilde{\beta}_2$ is the coefficient on $\tilde{W}_i$ in the population regression of Y_i on $\tilde{W}_i$. Thus:

$$\begin{aligned}\tilde{\beta}_2 &= \frac{Cov(Y_i, \tilde{W}_i)}{Var(\tilde{W}_i)} \\ &= \frac{Cov(Y_i, W_i)}{Var(W_i) + \sigma_u^2} = \beta_2 \frac{Var(W_i)}{Var(W_i) + \sigma_u^2} < \beta_2\end{aligned}$$

σ_u^2 generates “attenuation bias” and makes $\tilde{\beta}_2$ smaller (closer to zero) than β_2 .

3. [10 points] Now suppose that $Cov(X_i, W_i) = 1$, $Var(X_i) = 2$, and $Var(W_i) = 1$. Derive an explicit expression for β_1 in terms of the covariances of Y_i with X_i and W_i and the variance of X_i .

By FWL, we know that:

$$\beta_1 = \frac{Cov(Y_i, \tilde{X}_i)}{Var(\tilde{X}_i)}$$

where $\tilde{X}_i = X_i - W_i$ (since the coefficient on W_i in a regression of X_i on W_i is 1). Thus:

$$\beta_1 = \frac{Cov(Y_i, X_i - W_i)}{Var(X_i - W_i)} = Cov(Y_i, X_i) - Cov(Y_i, W_i)$$

4. [10 points] Continuing to make the same assumptions, derive an expression for $\tilde{\beta}_1$ and explain why it differs from β_1 .

Again by FWL, we know that:

$$\tilde{\beta}_1 = \frac{Cov(Y_i, X_i^*)}{Var(X_i^*)}$$

where $X_i^* = X_i - \frac{1}{1+\sigma_u^2}\tilde{W}_i$. Thus:

$$\tilde{\beta}_1 = \frac{Cov(Y_i, X_i - \frac{1}{1+\sigma_u^2}\tilde{W}_i)}{Var(X_i - \frac{1}{1+\sigma_u^2}\tilde{W}_i)} = \frac{Cov(Y_i, X_i) - \frac{1}{1+\sigma_u^2}Cov(Y_i, W_i)}{2 - \frac{1}{1+\sigma_u^2}}$$

It differs because even though there's no measurement error in X_i , the measurement error in $\tilde{W}_i$ still produces bias in $\tilde{\beta}_1$ for β_1 .

5. [5 points] Can you tell whether $\tilde{\beta}_1$ will be larger or smaller than β_1 in absolute value? How does this differ from the comparison between β_2 and $\tilde{\beta}_2$?

In general, you cannot tell whether $\tilde{\beta}_1$ will be attenuated towards zero. As an extreme case, let $\sigma_u^2 \rightarrow \infty$. Then:

$$\tilde{\beta}_1 \approx \frac{Cov(Y_i, X_i)}{2}$$

How this relates to β_1 depends on $Cov(Y_i, W_i)$. If that covariance is zero, then there will be attenuation towards zero, but if $Cov(Y_i, W_i)$ non zero, it is possible that $|\tilde{\beta}_1| > |\beta_1|$. It is even possible that $\tilde{\beta}_1$ and β_1 take different signs (e.g., if $Cov(Y_i, X_i)$ is negative and $CoV(Y_i, W_i)$ is large and positive.

6. [5 points] What does that tell you about how measurement error affects the interpretation of regressors in a multivariate regression?

In short, we know that measurement error in one variable attenuates its coefficient towards zero. When the (true) regressors are correlated, however, measurement error in one variable may make the coefficient on the variables either larger or smaller. So be careful!